

22. Armspan and Height Leonardo da Vinci (1452–1519) drew a sketch of a man, indicating that a person's armspan (measuring across the back with your arms outstretched to make a "T") is roughly equal to the person's height. To test this claim, we measured eight people with the following results:

Person	1	2	3	4
Armspan (centimeters)	172	158	165	176
Height (centimeters)	175	157	165	177

Person	5	6	7	8
Armspan (centimeters)	172	175	157	153
Height (centimeters)	170	170	160	157

- Draw a scatterplot for armspan and height. Use the same scale on both the horizontal and vertical axes. Describe the relationship between the two variables.
- If da Vinci is correct, and a person's armspan is roughly the same as the person's height, what should the slope of the regression line be?
- Calculate the regression line for predicting height based on a person's armspan. Does the value of the slope b confirm your conclusions in part b?
- If a person has an armspan of 157 centimeters, what would you predict the person's height to be?

c. $\hat{y} = a + bx$

$$b = \frac{\sum xy}{\sum x^2} = \frac{468}{568} = 0.824 \quad a = \bar{y} - b\bar{x} = 166.375 - 0.824 \times 166 = 29.591$$

$$\sum xy = r \cdot (n-1) \cdot s_x \cdot s_y = 468$$

$$\sum x^2 = (n-1) \cdot s_x^2 = 568$$

$$\hat{y} = 29.591 + 0.824x$$

d. $\hat{y} = 29.591 + 0.824 \times 157 = 158.959$